Chapter 3, Section 5

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June 14, 2026

1. Let X be a projective scheme over a field k, and let $\mathcal{F}$ be a coherent sheaf on X. We define the *Euler characteristic* of $\mathcal{F}$ by
$$\chi(\mathcal{F}) = \sum (-1)^i \dim_k \mathrm{H}^i(X, \mathcal{F}).$$
If
$$0 \longrightarrow \mathcal{F}' \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}'' \longrightarrow 0$$
is a short exact sequence of coherent sheaves on X, show that $\chi(\mathcal{F}) = \chi(\mathcal{F}') + \chi(\mathcal{F}'')$.

Proof. We get an exact sequence of finite dimensional k-vector spaces (5.2)
$$\cdots \xrightarrow{\delta_{i-1}} \mathrm{H}^i(X, \mathcal{F}') \longrightarrow \mathrm{H}^i(X, \mathcal{F}) \longrightarrow \mathrm{H}^i(X, \mathcal{F}'') \xrightarrow{\delta_i} \cdots,$$
where the δ_i's are connecting homomorphisms. By exactness, we have
$$\begin{aligned}
\chi(\mathcal{F}) &= \sum (-1)^i \dim_k \mathrm{H}^i(X, \mathcal{F}) \\
&= \sum (-1)^i (\dim_k \operatorname{coker} \delta_{i-1} + \dim_k \ker \delta_i) \\
&= \sum (-1)^i (\dim_k \mathrm{H}^i(X, \mathcal{F}') - \dim_k \operatorname{im} \delta_{i-1} + \dim_k \mathrm{H}^i(X, \mathcal{F}'') - \dim_k \operatorname{im} \delta_i) \\
&= \sum (-1)^i \dim_k \mathrm{H}^i(X, \mathcal{F}') + \sum (-1)^i \dim_k \mathrm{H}^i(X, \mathcal{F}'') \\
&= \chi(\mathcal{F}') + \chi(\mathcal{F}'').
\end{aligned}$$
$\square$

2. (a) Let X be a projective scheme over a field k, let $\mathcal{O}_X(1)$ be a very ample invertible sheaf on X over k, and let $\mathcal{F}$ be a coherent sheaf on X. Show that there is a polynomial $P(z) \in \mathbf{Q}[z]$, such that $\chi(\mathcal{F}(n)) = P(n)$ for all $n \in \mathbf{Z}$. We call P the *Hilbert polynomial* of $\mathcal{F}$ with respect to the sheaf $\mathcal{O}_X(1)$.

(b) Now let $X = \mathbf{P}^r_k$, and let $M = \Gamma_*(\mathcal{F})$, considered as a graded $S = k[x_0, \ldots, x_r]$-module. Use (5.2) to show that the Hilbert polynomial of $\mathcal{F}$ just defined is the same as the Hilbert polynomial of M defined in (I, §7).

Proof.

(a) Since $\mathcal{O}_X(1)$ is a very ample sheaf on X over k, there is a closed immersion $i : X \to \mathbf{P}^r_k$ of schemes over k, for some r, such that $\mathcal{O}_X(1) = i^*\mathcal{O}_{\mathbf{P}^r}(1)$. If $\mathcal{F}$ is coherent on X, then $i_*\mathcal{F}$ is coherent on $\mathbf{P}^r_k$ (II, Ex. 5.5), and the cohomology is the same (2.10). The Euler characteristic is defined in terms of cohomology, so we reduce to the case $X = \mathbf{P}^r_k$.
Cover X by a finite number of open affines $U_i \cong \operatorname{Spec} k[y_1, \ldots, y_r]$. Then there exist finitely generated $k[y_1, \ldots, y_r]$-modules M_i such that $\mathcal{F}\big|_{U_i} \cong \tilde{M}_i$. Since $\operatorname{Spec} A_i \cap \operatorname{Supp} \mathcal{F}$ is a closed subset of $\operatorname{Spec} A_i$; in fact, $\operatorname{Spec} A_i \cap \operatorname{Supp} \mathcal{F} = V(\operatorname{Ann} M_i)$ (AM, Ex. 3.19), $\operatorname{Supp} \mathcal{F}$ is a closed subset of X. We proceed by induction on $\dim \operatorname{Supp} \mathcal{F}$. If $\operatorname{Supp} \mathcal{F}$ has dimension zero, then $\operatorname{Supp} \mathcal{F} \cap U_i$ is affine (II, Ex. 3.11b) equal to spectra of a Noetherian ring A of dimension zero, which are precisely Artinian rings (AM, 8.5). So $U_i \cap \operatorname{Supp} \mathcal{F}$ is a finite set (AM, Ex. 8.2), and since X can be covered by a finite number of U_i, $\operatorname{Supp} \mathcal{F}$ is a finite set of closed points. Thus, $\mathcal{F}$ is isomorphic to a direct sum of skyscraper sheaves. More precisely, $\mathcal{F} = \bigoplus_{P \in \operatorname{Supp} \mathcal{F}} \mathcal{F}_P$, where $\mathcal{F}_P$ is the stalk of $\mathcal{F}$ at P, and cohomology commutes with direct sums, so by

(Ex. 1), we reduce to the case when $\operatorname{Supp}\mathcal{F}$ consists of a single point. Then, $\chi(\mathcal{F}(n))$ is just the Hilbert polynomial of $\mathcal{F}_P$ (I, 7.5) as a graded $A[y_1,\dots,y_r]$-module.

Assume the statement to be true for all coherent sheaves with support dimension less than n. Consider the exact sequence

$$0 \longrightarrow \mathcal{O}_X(-1) \xrightarrow{x_r} \mathcal{O}_X \longrightarrow \mathcal{O}_H \longrightarrow 0,$$

where H is the hyperplane $x_r = 0$. Taking the tensor product of this sequence with $\mathcal{F}$ gives us

$$0 \longrightarrow \mathcal{R} \longrightarrow \mathcal{F}(-1) \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}_H \longrightarrow 0,$$

where $\mathcal{R}$ is the kernel of the map $\mathcal{F}(-1) \to \mathcal{F}$. Twisting by $\mathcal{O}_X(n)$ is exact, so we get

$$\chi(\mathcal{F}(n)) - \chi(\mathcal{F}(n-1)) = \chi(\mathcal{F}_H(n)) - \chi(\mathcal{R}(n))$$

by (III, Ex. 5.1). Consider the coherent $\mathcal{O}_X$-module $\mathcal{R}$. It vanishes under the multiplication map x_r, where x_r locally defines one of the coordinates such that $x_r = 0$ is the defining equation for H, and such that there is a inclusion $\operatorname{Supp}\mathcal{R} \subseteq H$, which exists by Bertini's theorem. Since $H \cong \mathbf{P}_k^{r-1}$, we can apply the induction hypothesis to $\mathcal{R}$ and $\mathcal{F}_H$, so let $P_{\mathcal{R}}(z), P_{\mathcal{F}_H}(z) \in \mathbf{Q}[z]$ be polynomials such that

$$\chi(\mathcal{R}(n)) = P_{\mathcal{R}}(n), \quad \chi(\mathcal{F}_H(n)) = P_{\mathcal{F}_H}(n).$$

The difference function of $\chi(\mathcal{F}(n))$ is equal to $P_{\mathcal{F}_H}(z) - P_{\mathcal{R}}(n) \in \mathbf{Q}[z]$, so by (§1, 7.3), we conclude that $\chi(\mathcal{F}(n)) = P(n)$ for some $P(z) \in \mathbf{Q}[z]$.

(b) By (5.2), there exists an integer n_0 such that for each $i > 0$ and each $n \geq n_0$, $H^i(X, \mathcal{F}(n)) = 0$. Let $P(n)$ be the Hilbert polynomial just defined, and let $P'(n)$ be the polynomial from associated to the graded S-module M from (I, §7). Then $P(n) = \dim_k H^0(X, \mathcal{F}(n))$ for each $n \geq n_0$, but $M_n = \Gamma(X, \mathcal{F}(n)) = H^0(X, \mathcal{F}(n))$ and $P'(n) = \dim_k M_n$; in particular, $P(n)$ and $P'(n)$ agree at infinitely many values. Hence, they must be equal.

□

3. *Arithmetic Genus.* Let X be a projective scheme of dimension r over a field k. We define the *arithmetic genus* p_a of X by

$$p_a(X) = (-1)^r(\chi(\mathcal{O}_X) - 1).$$

Note that it depends only on X, not on any projective embedding.

(a) If X is integral, and k algebraically closed, show that $\mathrm{H}^0(X, \mathcal{O}_X) \cong k$, so that

$$p_a(X) = \sum_{i=0}^{r-1} (-1)^i \dim_k \mathrm{H}^{r-i}(X, \mathcal{O}_X).$$

In particular, if X is a curve, we have

$$p_a(X) = \dim_k \mathrm{H}^1(X, \mathcal{O}_X).$$

(b) If X is a closed subvariety of $\mathbf{P}_k^r$, show that this $p_a(X)$ coincides with the one defined in (I, Ex. 7.2), which apparently depended on the projective embedding.

(c) If X is a nonsingular projective curve over an algebraically closed field k, show that $p_a(X)$ is in fact a *birational* invariant. Conclude that a nonsingular plane curve of degree $d \geq 3$ is not rational.

Proof.

(a) For any embedding $j : X \to \mathbf{P}_k^r$, $\mathrm{H}^0(X, \mathcal{O}_X) \cong k$ (I, 3.4).

(b) Let $\mathcal{I}_X$ be the ideal sheaf of X on $\mathbf{P}_k^r$. The homogenous coordinate ring $S(X)$ of X for the given embedding is $k[x_0, \dots, x_r]/I$, where I is the ideal $\Gamma_*(\mathcal{I}_X)$ (II, Ex. 5.14). The result follows by (Ex. 1) and (Ex. 2b).

(c) Two nonsingular projective curves are birational if and only if they are isomorphic (II, 6.7), so they must have the same cohomology. If X is a nonsingular plane curve of degree $d \geq 3$, then

$$p_a(X) = \frac{1}{2}(d-1)(d-2) > 0$$

by (I, Ex. 7.2), but $p_a(\mathbf{P}^1) = 0$.

$\square$

6. *Curves on a Nonsingular Quadric Surface.* Let Q be the nonsingular quadric surface $xy = zw$ in $X = \mathbf{P}^3_k$ over a field k. We will consider locally principal closed subschemes Y of Q. These correspond to Cartier divisors on Q by (II, 6.17.1). On the other hand, we know that $\operatorname{Pic} Q \cong \mathbf{Z} \oplus \mathbf{Z}$, so we can talk about the *type* (a, b) of Y (II, 6.16) and (II, 6.6.1). Let us denote the invertible sheaf $\mathcal{L}(Y)$ by $\mathcal{O}_Q(a, b)$. Thus, for any $n \in \mathbf{Z}$, $\mathcal{O}_Q(n) = \mathcal{O}_Q(n, n)$.

(a) Use the special cases $(q, 0)$ and $(0, q)$, with $q > 0$, when Y is a disjoint union of q lines $\mathbf{P}^1$ in Q, to show:

  (1) if $|a - b| \leq 1$, then $H^1(Q, \mathcal{O}_Q(a, b)) = 0$;
  (2) if $a, b < 0$, then $H^1(Q, \mathcal{O}_Q(a, b)) = 0$;
  (3) if $a \leq -2$, then $H^1(Q, \mathcal{O}_Q(a, 0)) \neq 0$.

(b) Now use these results to show:

  (1) if Y is a locally principal closed subscheme of type (a, b) with $a, b > 0$, the Y is connected;
  (2) now assume k is algebraically closed. Then for any $a, b > 0$, there exists an irreducible nonsingular curve Y of type (a, b). Use (II, 7.6.2) and (II, 8.18).
  (3) an irreducible nonsingular curve Y of type (a, b), $a, b > 0$ on Q is projectively normal (II, Ex. 5.14) if and only if $|a - b| \leq 1$. In particular, this gives lots of examples of nonsingular, but not projectively normal curves in $\mathbf{P}^3$. The simplest is the one of type $(1, 3)$, which is just the rational quartic curve (I, Ex. 3.18).

(c) If Y is a locally principal subscheme of type (a, b) in Q, show that $p_a(Y) = ab - a - b + 1 = (a-1)(b-1)$.

Proof.

(a)

(b) (1)

(2)

(3)

(c)

$\square$

7. Let X (respectively, Y) be proper schemes over a Noetherian ring A. We note by $\mathcal{L}$ an invertible sheaf.

(a) If $\mathcal{L}$ is ample on X, and Y is any closed subscheme of X, then $i^*\mathcal{L}$ is ample on Y, where $i : Y \to X$ is the inclusion.

(b) $\mathcal{L}$ is ample on X if and only if $\mathcal{L}_{\mathrm{red}} = \mathcal{L} \otimes \mathcal{O}_{X_{\mathrm{red}}}$ is ample on X_{red}.

(c) Suppose X is reduced. Then $\mathcal{L}$ is ample on X if and only if $\mathcal{L} \otimes \mathcal{O}_{X_i}$ is ample on X_i, for each irreducible component X_i of X.

(d) Let $f : X \to Y$ be a finite surjective morphism, and let $\mathcal{L}$ be an invertible sheaf on Y. Then $\mathcal{L}$ is ample on Y if and only if $f^*\mathcal{L}$ is ample on X.

Proof.

(a) If $\mathcal{F}$ is any coherent sheaf on Y, then $f_*\mathcal{F}$ is a coherent $\mathcal{O}_X$-module (II, 5.5c)

(b)

(c)

(d) By (b) and (c), we reduce to the case X and Y are irreducible, reduced, and proper schemes over A.

$\square$

10. Let X be a projective scheme over a Noetherian ring A, and let $\mathcal{F}^1 \to \mathcal{F}^2 \to \cdots \to \mathcal{F}^r$ be an exact sequence of coherent sheaves on X. Show that there is an integer n_0, such that for all $n \geq n_0$, the sequence of global sections

$$\Gamma(X, \mathcal{F}^1(n)) \longrightarrow \Gamma(X, \mathcal{F}^2(n)) \longrightarrow \cdots \longrightarrow \Gamma(X, \mathcal{F}^r(n))$$

is exact.

Proof. Denote $f_i : \mathcal{F}^i :\to \mathcal{F}^{i+1}$ for each $i = 1, \ldots, n-1$. For every $i = 2, \ldots, n-1$, we have the following three short exact sequence

$$0 \longrightarrow \operatorname{im} f_{i-1} \longrightarrow \mathcal{F}^i \longrightarrow \operatorname{im} f_i \longrightarrow 0,$$

$$0 \longrightarrow \ker f_{i-1} \longrightarrow \mathcal{F}^{i-1} \longrightarrow \operatorname{im} f_{i-1} \longrightarrow 0,$$

$$0 \longrightarrow \operatorname{im} f_i \longrightarrow \mathcal{F}^{i+1} \longrightarrow \operatorname{coker} f_i \longrightarrow 0.$$

Every sheaf above is coherent (II, 5.7). By (5.2), there exists an integer n_i such that for each $i > 0$ and each $n \geq n_i$, the cohomology vanishes for every sheaf, i.e., there is an exact sequence of global sections

$$0 \longrightarrow \Gamma(X, \operatorname{im} f_{i-1}(n)) \longrightarrow \Gamma(X, \mathcal{F}^i(n)) \longrightarrow \Gamma(X, \operatorname{im} f_i(n)) \longrightarrow 0$$

$$0 \longrightarrow \Gamma(X, \ker f_{i-1}(n)) \longrightarrow \Gamma(X, \mathcal{F}^{i-1}(n)) \longrightarrow \Gamma(X, \operatorname{im} f_{i-1}(n)) \longrightarrow 0$$

$$0 \longrightarrow \Gamma(X, \operatorname{im} f_i(n)) \longrightarrow \Gamma(X, \mathcal{F}^{i+1}(n)) \longrightarrow \Gamma(X, \operatorname{coker} f_i(n)) \longrightarrow 0$$

which implies the sequence

$$\Gamma(X, \mathcal{F}^{i-1}(n)) \longrightarrow \Gamma(X, \mathcal{F}^i(n)) \longrightarrow \Gamma(X, \mathcal{F}^{i+1}(n))$$

is exact. Take $n_0 = \max n_i$ to obtain the desired integer. $\square$